Printable photography reference cards

ShutterMath
Field Kit

Six field-ready cheat cards — depth of field, exposure, ND filters, hyperfocal distance, astro star exposure, and print resolution. All values computed with the standard formulas, checked, and formatted for your camera bag.

6 cards · A4 landscape

Print at home

Free lifetime updates

shuttermath.com — free tools, field-ready cards

Depth of Field

Near limit, far limit and hyperfocal — computed for your exact sensor.

Formulas

$H = f^2 / (N \cdot c) + f$
$near = H \cdot s / (H + s - f)$ · $far = H \cdot s / (H - s - f)$

Circle of confusion (diagonal ÷ 1500)

Format	CoC (mm)
Full frame	0.029
APS-C (Sony/Nikon/Fuji)	0.019
Canon APS-C	0.018
Micro 4/3	0.014

50 mm @ f/2.8, full frame

Focus	Near	Far
2.00 m	1.88 m	2.13 m
3.00 m	2.74 m	3.31 m
5.00 m	4.31 m	5.95 m
10 m	7.57 m	15 m

Rules of thumb

- Doubling the f-number doubles the DoF.
- Doubling the focus distance ~doubles the DoF.
- Halving the focal length roughly quadruples DoF.
- Focus at H and everything from H/2 to ∞ is sharp.

Worked example

Portrait at 85 mm f/1.8, 2 m: near 1.97 m, far 2.03 m — about 5.5 cm of sharpness. That's why eyes are sharp and ears are soft. For a group shot at 35 mm f/4, 3 m: near ~2.5 m to ~3.8 m — enough for one row, not three.

Landscape: 24 mm f/8 on full frame → H ≈ 2.5 m. Focus at 2.5 m and everything from 1.25 m to infinity is sharp.

Exposure

EV, the exposure triangle, and the stops that tie them together.

Formulas

$EV = \log_2(N^2 / t)$ (at ISO 100)
$t_2 = t_1 \times 2^{stops}$ · $N_2 = N_1 \times \sqrt{2}^{stops}$

f-stop light scale

f/	1	1.4	2	2.8	4	5.6	8	11	16
light	1×	½	¼	⅛	1/16	1/32	1/64	1/128	1/256

Scene EV reference (ISO 100)

EV	Scene	Aperture @ 1/125s
-2	Full-moon night	— @1/125s
0	Very dim interior	— @1/125s
3	City night	— @1/125s
6	Overcast evening	— @1/125s
9	Bright overcast	f/2.0 @1/125s
12	Bright daylight	f/5.7 @1/125s
15	Sunny 16	f/16.2 @1/125s

Sunny 16: bright sun → f/16 at 1/ISO s. Overcast → f/8. Heavy shade → f/5.6.

Worked example

f/2.8, 1/250s, ISO 100 → EV ≈ 11. Want f/4 (one stop less light)? Shutter drops to 1/125s to compensate. Want to freeze at 1/1000s? That's two stops — open to f/2 or raise ISO to 400.

The error to never make: change one setting and forget to compensate the other. Aperture, shutter and ISO always move in pairs.

ND Filter

Exact shutter times for neutral density filters — and the 180° video rule.

Formula

result = base × 2^stops

Filter strength

Filter	Factor	Stops
ND2	2×	1 stops
ND4	4×	2 stops
ND8	8×	3 stops
ND16	16×	4 stops
ND64	64×	6 stops
ND256	256×	8 stops
ND1000	1000×	10 stops
ND100000	100000×	17 stops

Base exposure → result

Base	ND64 (6 stops)	ND1000 (10 stops)
1/1000s	1/16 s	1 s
1/500s	1/8 s	2 s
1/250s	1/4 s	4 s
1/125s	1/2 s	8 s
1/60s	1.06667 s	16.6667 s
1/30s	2.13333 s	33.3333 s

Patterns: every 3 stops ≈ 8× the time. ND64 = "moving water at dusk", ND1000 = "cotton-candy water at midday".

180° video rule

fps	Shutter
24	1/48 s
30	1/60 s
60	1/120 s
120	1/240 s

Worked example: daylight base 1/1000s at 24 fps needs shutter 1/48s = 4.4 stops of light removed → grab an ND16 (4 stops) and fine-tune with aperture.

Hyperfocal

Focus at H and everything from half of H to infinity is sharp.

Formula

$H = f^2 / (N \cdot c) + f$
The rule: set your lens to H (or a click past it) and the entire scene from H/2 to infinity is acceptably sharp. Focus closer → lose the horizon. Focus farther → waste depth in front.

Hyperfocal distance @ f/8 (metres)

Format	16mm	24mm	35mm	50mm
Full frame	1.13 m	2.52 m	5.35 m	11 m
APS-C (Sony/Nikon/Fuji)	1.72 m	3.85 m	8.18 m	17 m
Micro 4/3	2.24 m	5.02 m	11 m	22 m

Reading it: 24 mm on full frame at f/8 → H ≈ 2.5 m; everything from 1.25 m to infinity sharp. On Micro 4/3, the same lens needs ≈ 1.1 m (2× crop ≈ 2× more depth).

Worked example

Landscape: 24 mm, f/8, full frame → focus at 2.5 m. Foreground rock at 1.5 m sharp, mountain at ∞ sharp.

Focus stacking: when H/2 still isn't close enough, shoot at the nearest point you need, then step focus toward H — two frames at f/8 usually beat one at f/16 (diffraction).

Astro — Star Exposure

Sharp stars: the 500/400/300 rules and the pixel-accurate NPF rule.

Rules (full-frame-equivalent focal length)

$t = 500 / f$ · $t = 400 / f$ · $t = 300 / f$

NPF rule (pixel-accurate)

$t = (35 \cdot N + 30 \cdot p) / f$ (p = pixel size, µm)
Example (A1 II, 4.17 µm): 20 mm f/2 → 500-rule 25 s, NPF 9.8 s. Modern high-res sensors streak long before the 500 rule says so.

Max shutter at f/2, full frame

Focal	500 rule	NPF
14 mm	36 s	14 s
20 mm	25 s	10 s
24 mm	21 s	8 s
35 mm	14 s	6 s

Pixel scale

arcsec/px = 206.3 × p(µm) / f(mm)
1–2″/px is the sharp-stars sweet spot; above 3″/px you're undersampling.

Worked example

Milky Way: 24 mm f/2 on full frame → 500 rule 20 s, NPF ≈ 11 s. Shoot 11 s, ISO 3200, stack 5–10 frames for the clean version.

Rule of thumb: wide + fast beats long. A 14 mm f/2.8 at 30 s shows trails on a 45 MP body — drop to 20 s or open up.

Print Resolution

Megapixels, DPI and print size — how big can you actually print?

Formulas

print size = pixels / DPI
pixels needed = inches × DPI

What DPI do you need?

Viewing	DPI
Gallery, close look	300
Books / magazines	240–300
Wall art, 1 m away	150–200
Billboard, 5 m away	30–60

Max print size (3:2 cameras)

Camera	@150	@240	@300
12 MP	28 × 19 in	18 × 12 in	14 × 9 in
24 MP	40 × 27 in	25 × 17 in	20 × 13 in
45 MP	55 × 37 in	34 × 23 in	27 × 18 in
61 MP	64 × 43 in	40 × 27 in	32 × 21 in

Worked example

24 MP (6000 × 4000): 20 × 13 in at 300 DPI, 25 × 17 in at 240, 40 × 27 in at 150. A 24″ wide gallery print at 250 DPI is comfortably in range.

Upscaling is fine: modern editors (and AI upscalers) handle +50% for large prints — don't fear a 45 MP file at 48″ wide.